

- Q.21 Compare BJT and FET transistors. (CO6)
Q.22 What are filters in rectifiers? Explain their types. (CO7)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Describe how a single-phase transformer works and what it is used for. (CO4)
Q.24 Explain the different types of electrical machines, focussing on DC generators and motors. (CO5)
Q.25 Discuss the basics of digital and analog principles, including operational amplifiers and logic gates. (CO7)

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3th Year / Advance Diploma in Tool & Die Making Subject : Basics of Electrical & Electronics Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What is the RMS value of an AC current? (CO1)
a) Average value b) Maximum value
c) Effective value d) Peak value
Q.2 Which law says that the total current entering a point is equal to the total current leaving? (CO2)
a) Ohm's Law
b) Kirchhoff's Current Law
c) Faraday's Law
d) Lenz's Law
Q.3 The induced EMF in a coil is caused by : (CO3)
a) A constant magnetic field
b) A changing magnetic field
c) Ohm's Law
d) Kirchhoff's voltage Law

- Q.4 A transformer works on the principle of. (CO4)
- Self-Induction
 - Mutual induction
 - Electromagnetic induction
 - All of the above
- Q.5 What does a DC motor do? (CO4)
- Converts mechanical energy to electrical energy
 - Always runs at a constant speed
 - Converts electrical energy to mechanical energy
 - Cannot reverse direction
- Q.6 A diode allows current to flow: (CO5)
- In both directions
 - Only in reverse direction
 - Only in forward direction
 - In neither direction

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The peak value of an AC waveform is _____ the RMS value. (CO1)

- Q.8 The energy stored in a magnetic field depends on _____ and _____. (CO3)
- Q.9 A three-phase induction motor requires _____ phase of power. (CO4)
- Q.10 IN a PN junction diode, the main charge carriers in the P-Type region are _____. (CO5)
- Q.11 Back EMF in a DC motor opposes the _____ voltage. (CO3)
- Q.12 A half-wave rectifier allows currents to flow during _____ half of the AC cycle. (CO7)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What is the difference between AC and DC circuits? (CO1)
- Q.14 Explain Ohm's Law with a simple example. (CO2)
- Q.15 How does a transformer work? (CO4)
- Q.16 What is electromagnetic induction? Why is it important? (CO3)
- Q.17 Describe how a DC motor works. (CO5)
- Q.18 What are the uses of induction motors? (CO5)
- Q.19 Explain the working of a PN junction diode. (CO6)
- Q.20 What is a stepper motor? How does it work? (CO6)